

Introduction for genetic material

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DNA Definition

- ***DNA is a group of molecules that is responsible for carrying and transmitting the hereditary materials or the genetic instructions from parents to offspring.***
- DNA is known as Deoxyribonucleic Acid. It is an organic compound that has a unique molecular structure. It is found in all prokaryotic cells , eukaryotic cells and many viruses.
- ✓ ***Nuclear DNA*** is the DNA contained within the nucleus of every cell in a eukaryotic organism.
- ✓ The DNA present in the mitochondria of the cell is termed ***mitochondrial DNA***. It is inherited from the mother to the child. In humans, there are approximately 16,000 base pairs of mitochondrial DNA.

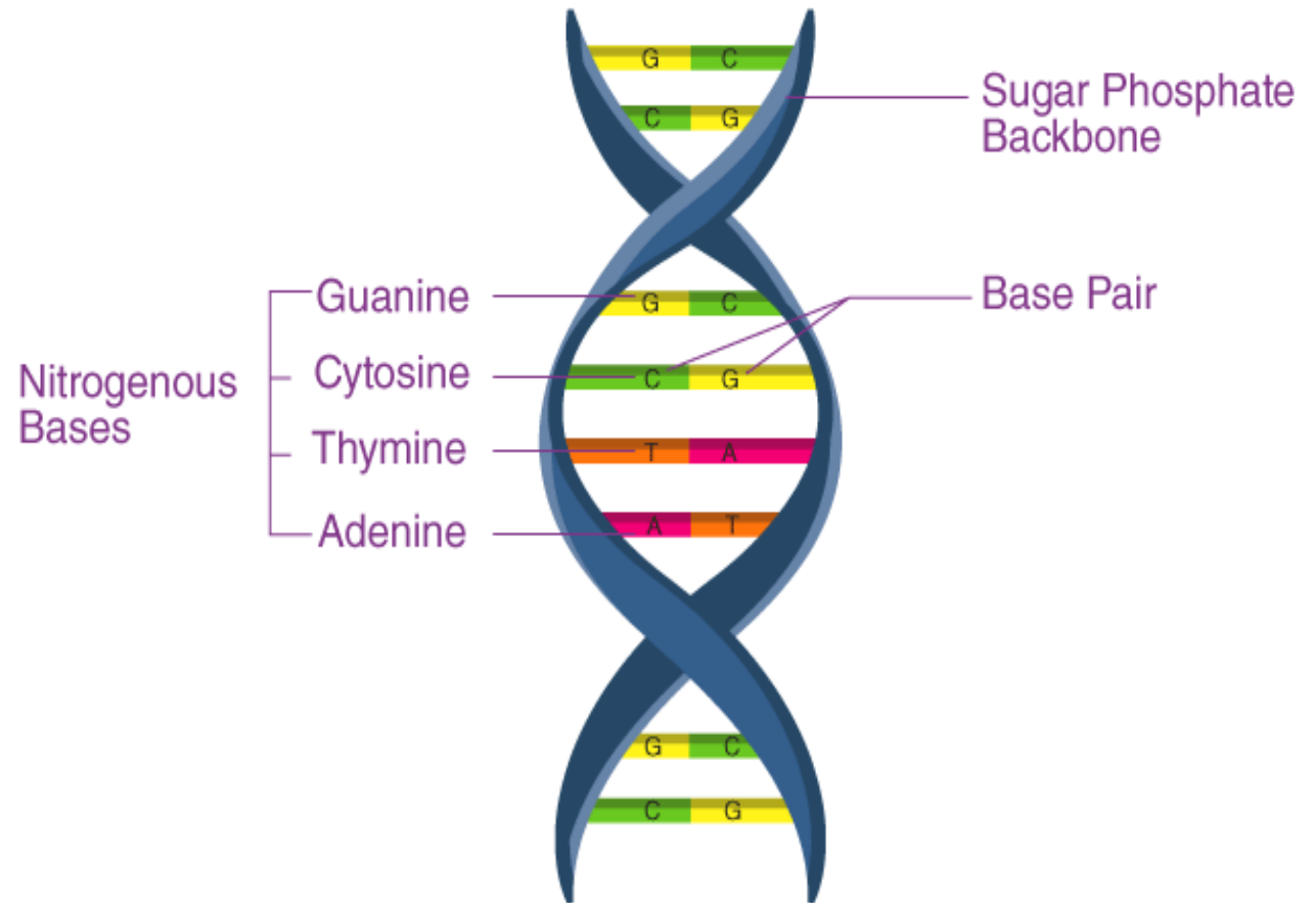
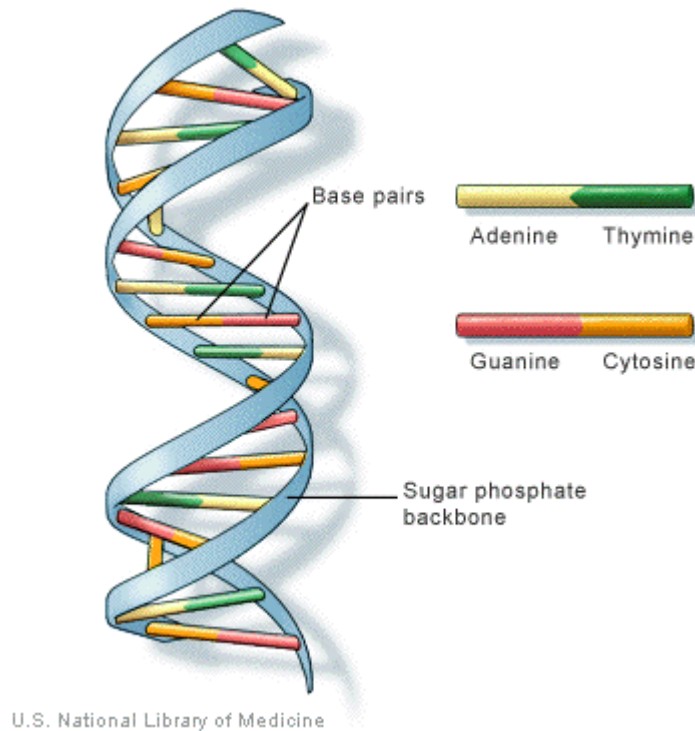
- In ***prokaryotes***, which do not have a membrane-bound nucleus, the DNA is found as a single circular extrachromosomal DNA known as ***plasmids***, which are *autonomous, self-replicating* genetic material in the cytoplasm.
- Some prokaryotes, such as bacteria, have plasmids have been used extensively in recombinant DNA technology to study gene expression.
- The configuration of the DNA molecule is highly stable, allowing it to act as a template for the ***replication*** of new DNA molecules, as well as for the production (***transcription***) of the related ***RNA*** (ribonucleic acid) molecule. A segment of DNA that codes for the cell's synthesis of a specific ***protein*** is called a ***gene***.
- A gene contains the basic three regions, *promoter*, *coding sequence* (reading frame), and terminator.

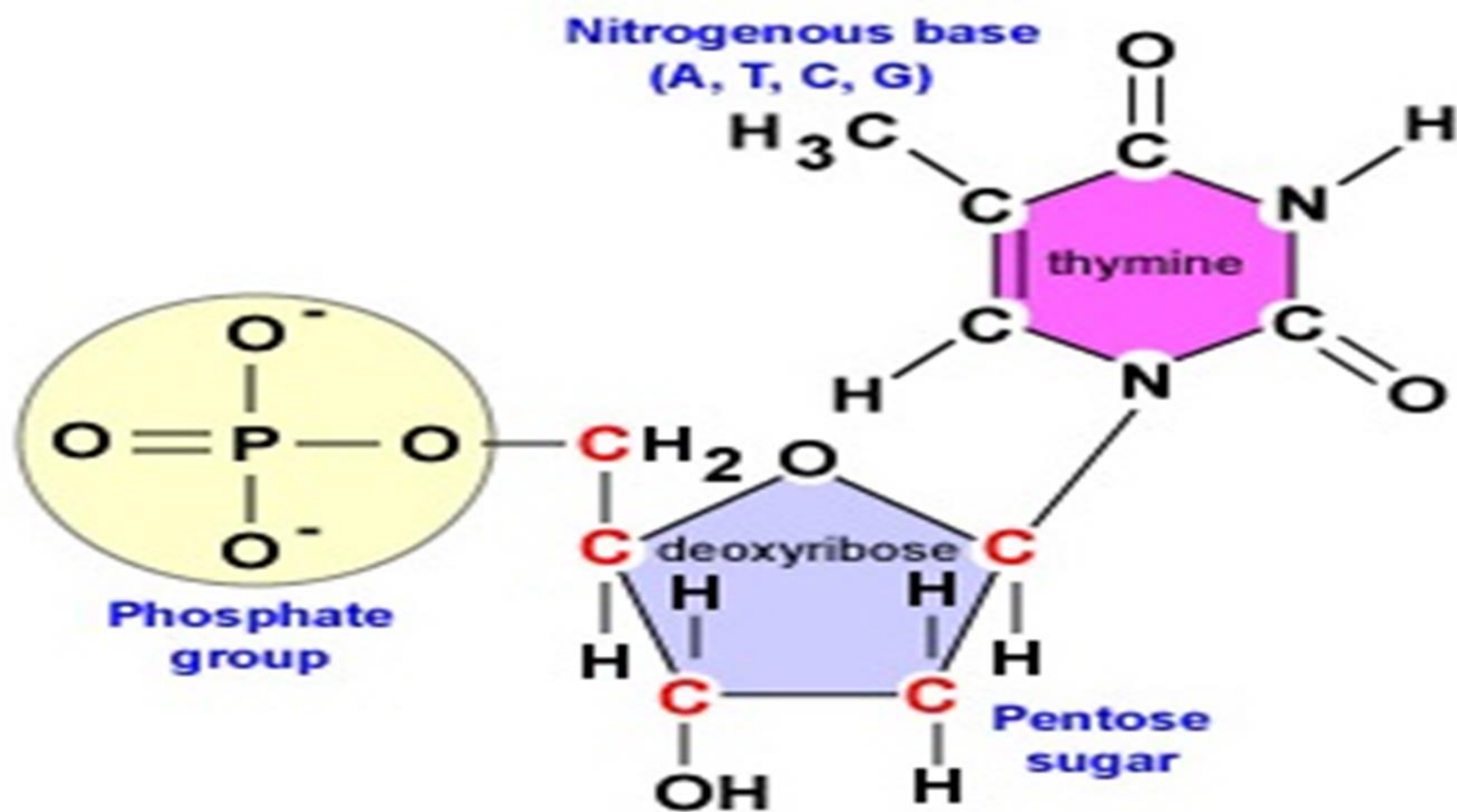
DNA structure

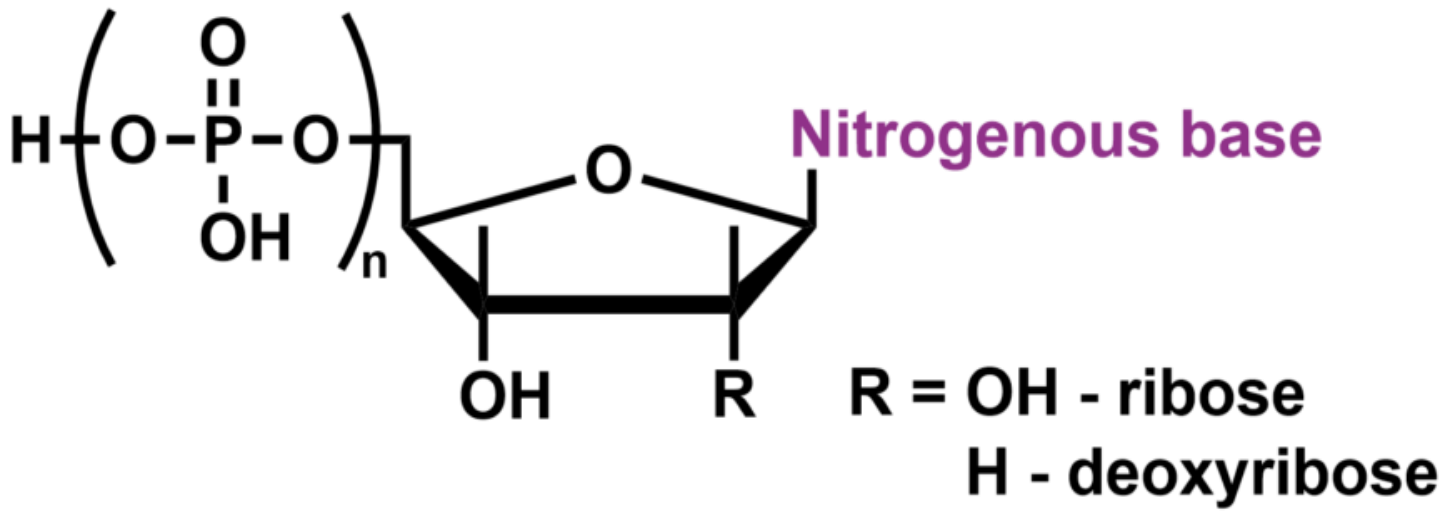
- The DNA structure can be thought of as a twisted ladder. This structure is described as a double-helix.
- The basic building blocks of DNA are nucleotides (polynucleotides), which are composed of a sugar group, a phosphate group, and a nitrogen base. The sugar and phosphate groups link the nucleotides together to form each strand of DNA. Adenine (A), Thymine (T), Guanine (G) and Cytosine (C) are four types of nitrogen bases.
- The order of the nitrogenous bases determines the genetic code or the DNA's instructions.

The following diagram explains the DNA structure representing the different parts of the DNA. DNA comprises a sugar-phosphate backbone and the nucleotide bases (guanine, cytosine, adenine and thymine).

DNA STRUCTURE







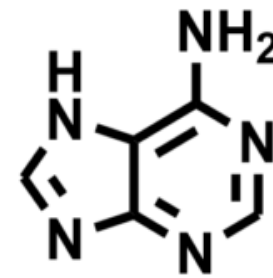
Nucleoside

Nucleotide

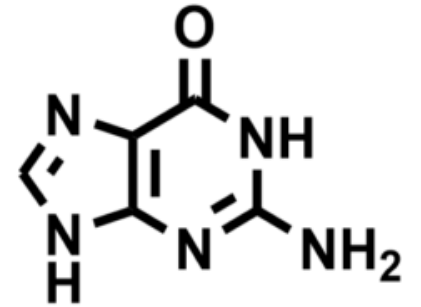
- n = 1 - nucleoside monophosphate
- 2 - nucleoside diphosphate
- 3 - nucleoside triphosphate

Nitrogenous bases:

Purines

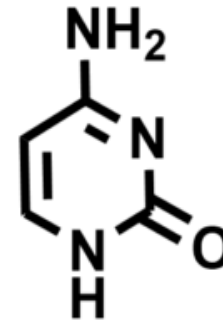


Adenine

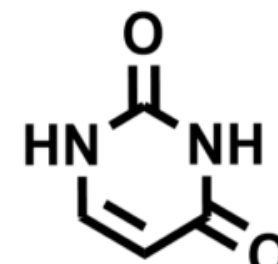


Guanine

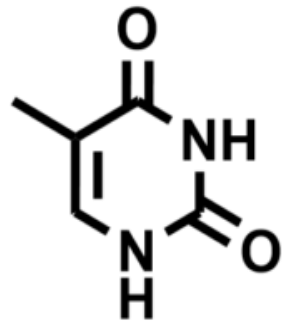
Pyrimidines



Cytosine



Uracil

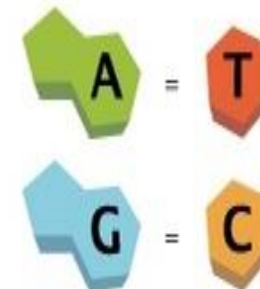


Thymine

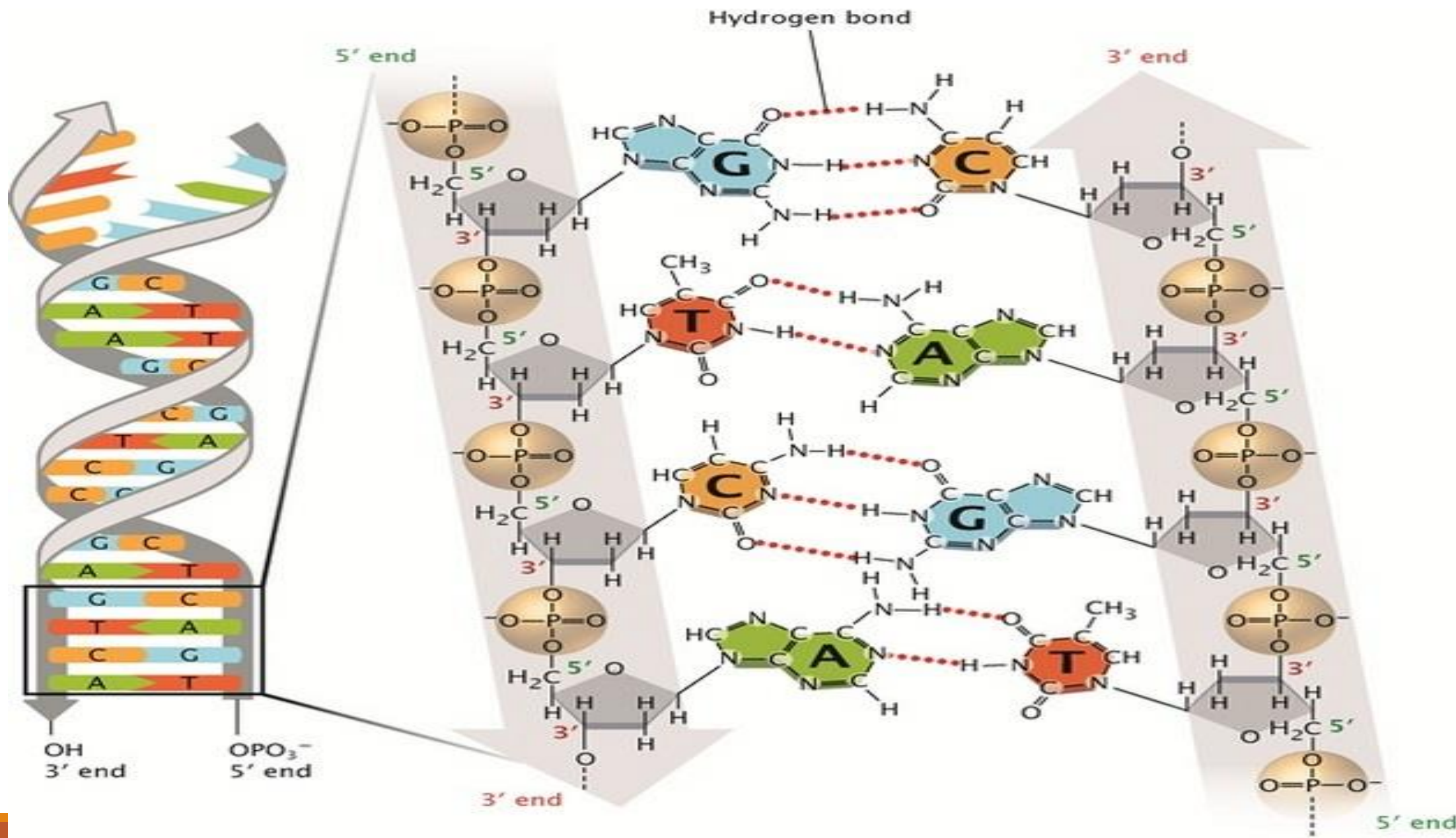
Erwin Chargaff rule

- **Erwin Chargaff**, a biochemist, discovered that the number of **nitrogenous bases in the DNA** was present in equal quantities. The amount of A is equal to T, whereas the amount of C is equal to G.
- In other words, the DNA of any cell from any organism should have a 1:1 ratio of purine and pyrimidine bases.

Total Purines = Total Pyrimidines



- The two strands of DNA run in opposite directions. These strands are held together by the **hydrogen bond** that is present between the two complementary nitrogenous bases(forming a ladder-like structure).
- The strands are helically twisted, where each strand forms a right-handed coil, and ten nucleotides make up a single turn.
- The nucleotides of each strand are bound to each other by covalent 3(C-3-OH group of sugar) -5(C-5-OH group of sugar of next nucleotide) by **phosphodiester linkage**.
- The DNA coils up, forming **chromosomes**, and each chromosome has a single molecule of DNA in it.



Chromosome Definition

A *chromosome* is a DNA molecule that consists of a part or all of the genetic material of an organism.

- The chromosome is present in the nucleus of each cell, and it is packaged into thread-like structures. Typically, chromosomes are not visible under the microscope, the only time they are visible is during the process of cell division.
- All living organisms have chromosomes; humans have 23 pairs of chromosomes. 22 pairs of these chromosomes are called *autosomes* – these are identical in males and females. The 23rd pair of chromosomes are called *allosomes* and they differ between the sexes. Males have one “X” and “Y” chromosome, while females have two copies of the “X” chromosome.



1 Cell

2 Nucleus

3 Chromosome

4 DNA

Chromatin Definition

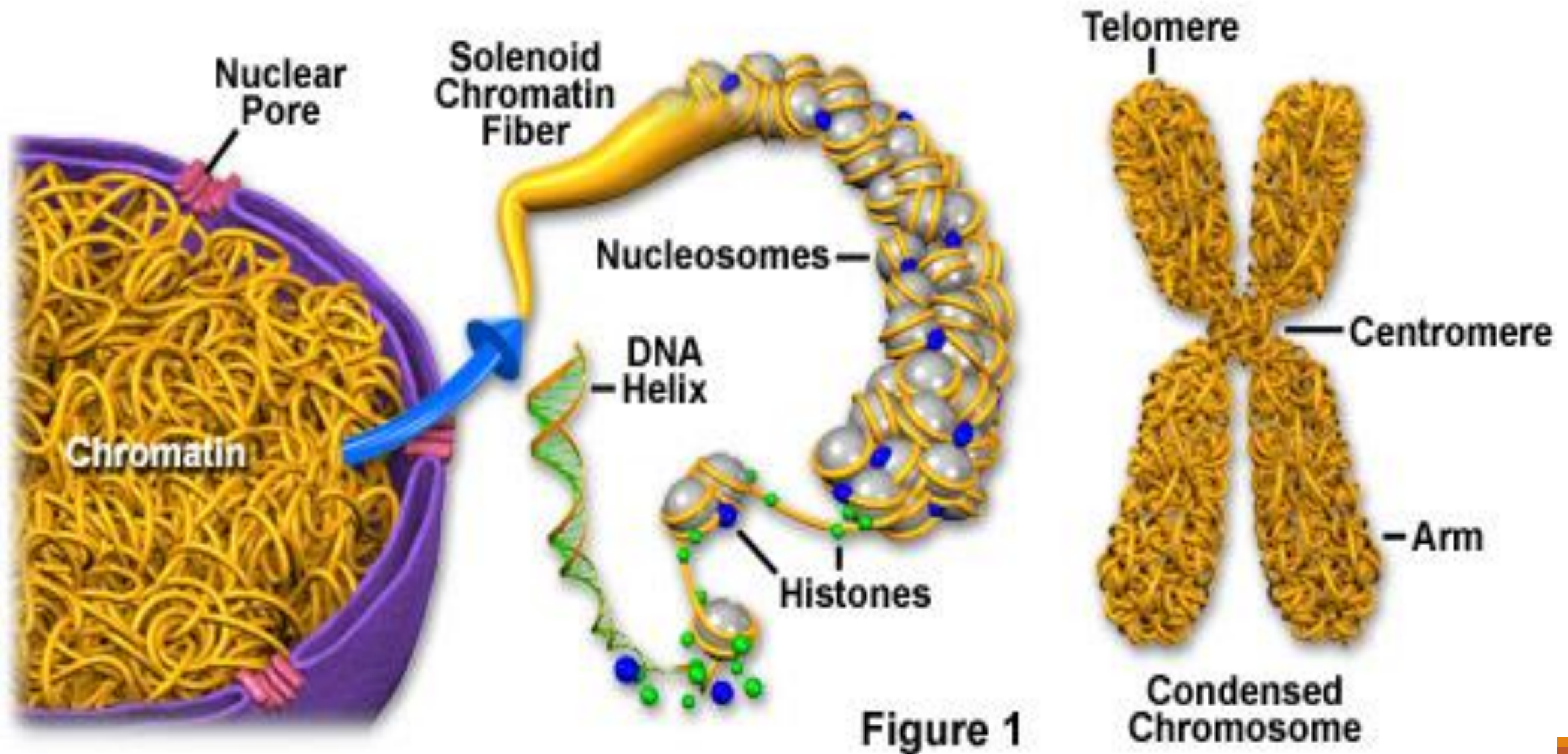
- Each DNA strand wraps around groups of small protein molecules called **histones**, forming a series of bead-like structures, called **nucleosomes**, connected by the DNA strand.
- The string of nucleosomes, is then coiled into an even denser structure known as a **solenoid**. The solenoid structure then coils to form a hollow tube.
- *This complex compression and structuring of DNA serves several functions:*
 - ✓ The overall negative charge of the DNA is neutralized by the positive charge of the histone molecules.
 - ✓ The DNA takes up much less space, and inactive DNA can be folded until it is needed.

Chromatin types

1-Euchromatin is the genetically *active* type of chromatin involved in transcribing RNA to produce proteins used in cell function and growth. The predominant type of chromatin found in cells during interphase, euchromatin is more diffuse than the other kind of chromatin, which is termed as heterochromatin.

2-Heterochromatin the additional compression of heterochromatin is thought to involve various proteins in addition to the histones, and the DNA it contains is thought to be genetically *inactive*. Heterochromatin tends to be most concentrated along chromosomes at certain regions of the structures, such as the **centromeres** and **telomeres**.

Chromatin and Condensed Chromosome Structure



RNA Definition

- ❑ **Ribonucleic acid (RNA)** is a type of molecule that consists of a long chain of nucleotide units. Each nucleotide consists of a nitrogenous base, a ribose sugar, and a phosphate.
- RNA is very similar to DNA, but differs in a few important structural details: in the cell,
 - ✓ RNA is usually ***single-stranded***, while DNA is usually ***double-stranded***.
 - ✓ RNA nucleotides *contain **ribose*** while DNA contains ***deoxyribose*** (a type of ribose that lacks one oxygen atom).
 - ✓ and RNA *has the base **uracil*** rather than ***thymine*** that is present in DNA.



DNA vs. RNA

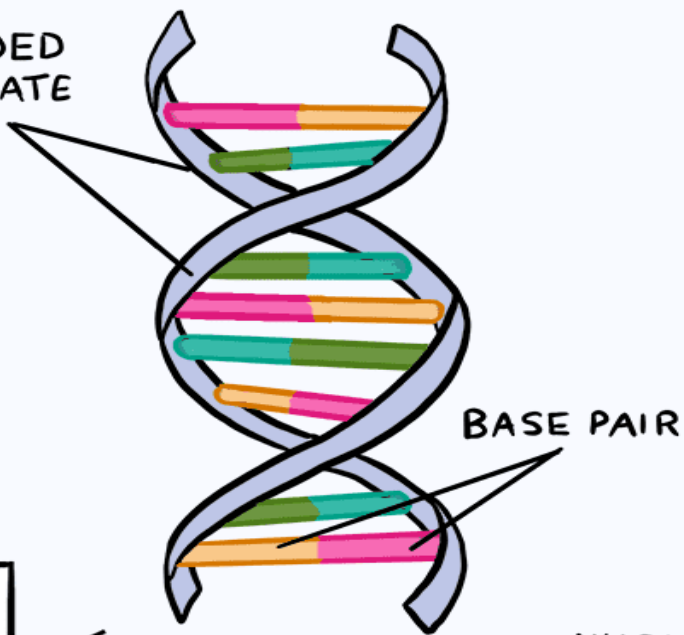
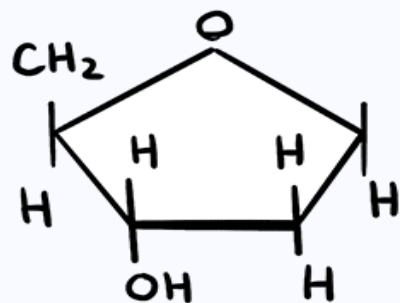


DEOXYRIBONUCLEIC ACID

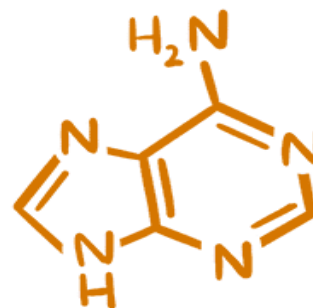
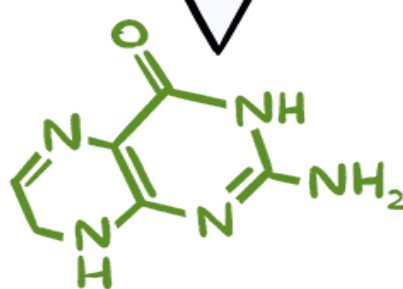
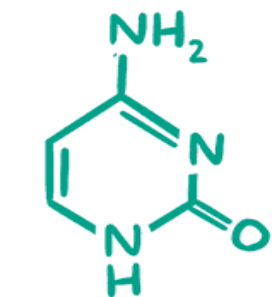
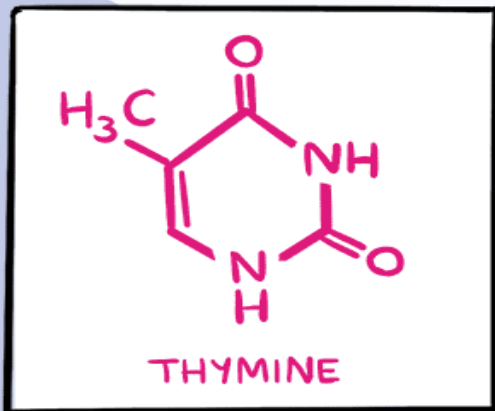
RIBONUCLEIC ACID

DOUBLE-STRANDED
SUGAR*PHOSPHATE

* DEOXYRIBOSE

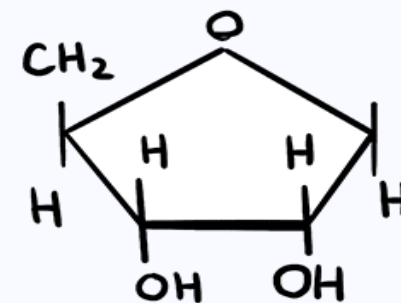


NUCLEOBASES

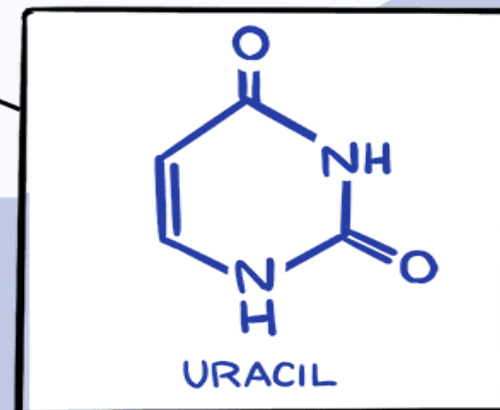


USUALLY SINGLE-STRANDED
SUGAR*PHOSPHATE

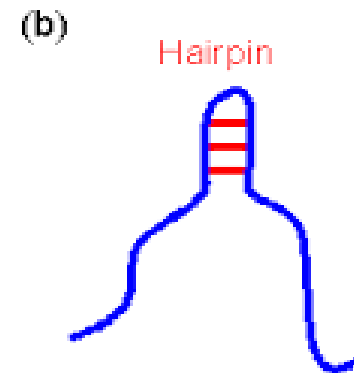
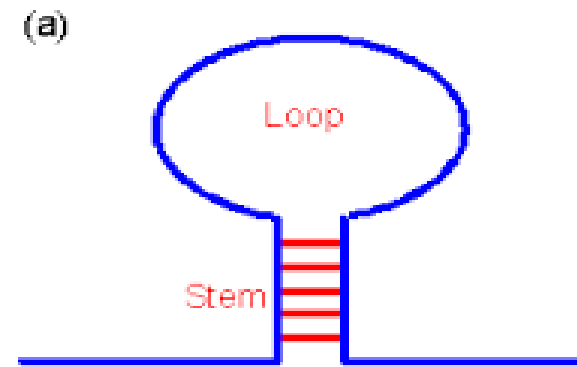
★ RIBOSE



SINGLE
NUCLEOBASE



- Most cellular RNA *is single stranded*, although some viruses have double stranded RNA(e.g. retrovirus).
- The single RNA strand is folded upon itself, either entirely or in certain regions. In the folded region a majority of the bases are complementary and are joined by hydrogen bonds. This helps in the stability of the molecule, such as stem-loop and hairpin (secondary structures).
- In regions where purine-pyrimidine pairing takes place, adenine pairs with uracil and guanine with cytosine.
- In the unfolded region the bases have no complements.



Types of RNA

1-Messenger RNA (mRNA) or template RNA

2-Ribosomal RNA (rRNA) and

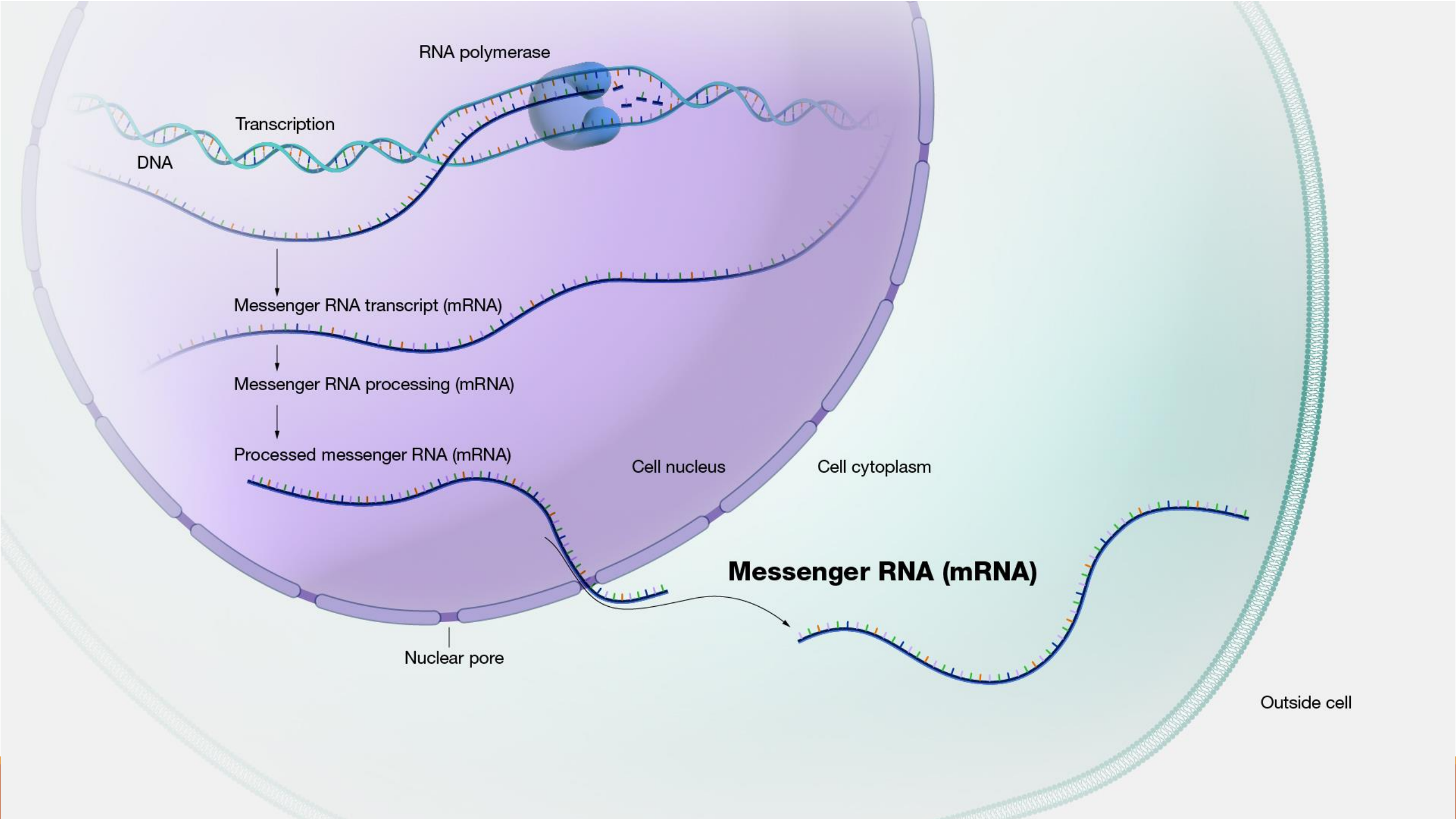
3-Soluble RNA (sRNA) or transfer RNA (tRNA)

4-More RNA molecule like enzymes and microRNAs.

Messenger RNA - mRNA

- Jacob and Monod (1961) proposed the name messenger RNA for the RNA carrying information for protein synthesis from the DNA (genes) in the nucleus by transcription process to the sites of protein formation (ribosomes) in the cytoplasm of the cell by translation process.
- It consists of only 3 to 5% of the total cellular RNA.
- The cell does not contain large quantities of mRNA. This is because mRNA, unlike other RNAs is constantly undergoing breakdown. It is broken down to its constituent ribonucleotides by ribonucleases enzyme.
- There is no base pairing. In fact base pairing in the mRNA strand destroys its biological activity.
- Since mRNA is transcribed on DNA (genes), its base sequence is complementary to that of the segment of DNA on which it is transcribed.

- When a coding region of mRNA codes for a single protein, it is known as *monocistronic* as in most eukaryotic mRNAs. In prokaryotes, most mRNAs code for more than one protein, it is called *polycistronic*. These proteins mostly have related functions and are regulated by a single regulatory region having *promoter* and *operator regions*. In humans, the mitochondrial genome is polycistronic.
- mRNA directs a cell to synthesize a protein. The administration of modified mRNA sequence can be utilized to cure diseases and also can be used as a vaccine. Several mRNA vaccines are developed and are being worked upon to protect against COVID-19 infection and have got approval for restricted usage, e.g. Pfizer-BioNTech and the Moderna vaccine.



Structural Features of mRNA

1. Cap. At the 5' end of the mRNA molecule in most eukaryote cells and animal virus molecules is found a 'cap'. This is blocked methylated structure(7-methylguanosine triphosphate).

- The rate of protein synthesis depends upon the presence of the cap. Without the cap mRNA molecules bind very poorly to the ribosomes ,and it helps in stabilizing the mRNA by preventing the attack of 5'-*exonucleases* enzymes.

2. Noncoding region 1 (NC1). The cap is followed by a region of 10 to 100 nucleotides. This region is rich in A and U residues, and does not translate protein.

3. The initiation codon(start codon) is AUG in both prokaryotes and eukaryotes

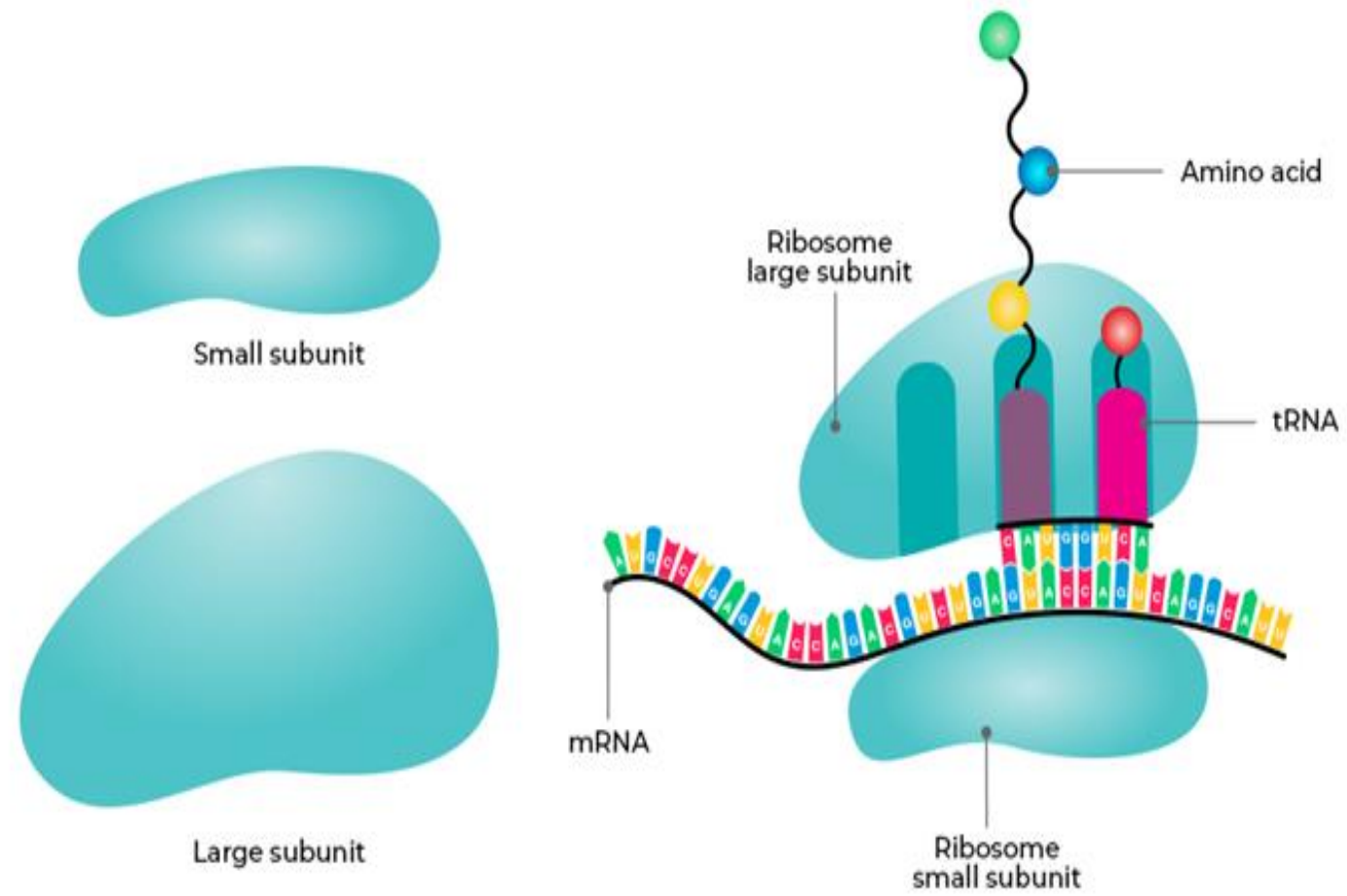
4. The coding region consists of about 1,500 nucleotides on the average and translates protein. The coding region begins with the start codon, i.e. 'AUG' and ends with any of the stop codons, i.e. UAG, UAA or UGA.

5. Poly(A) Tail – Polyadenylate tail is present at the 3'-end.



Ribosomal RNA

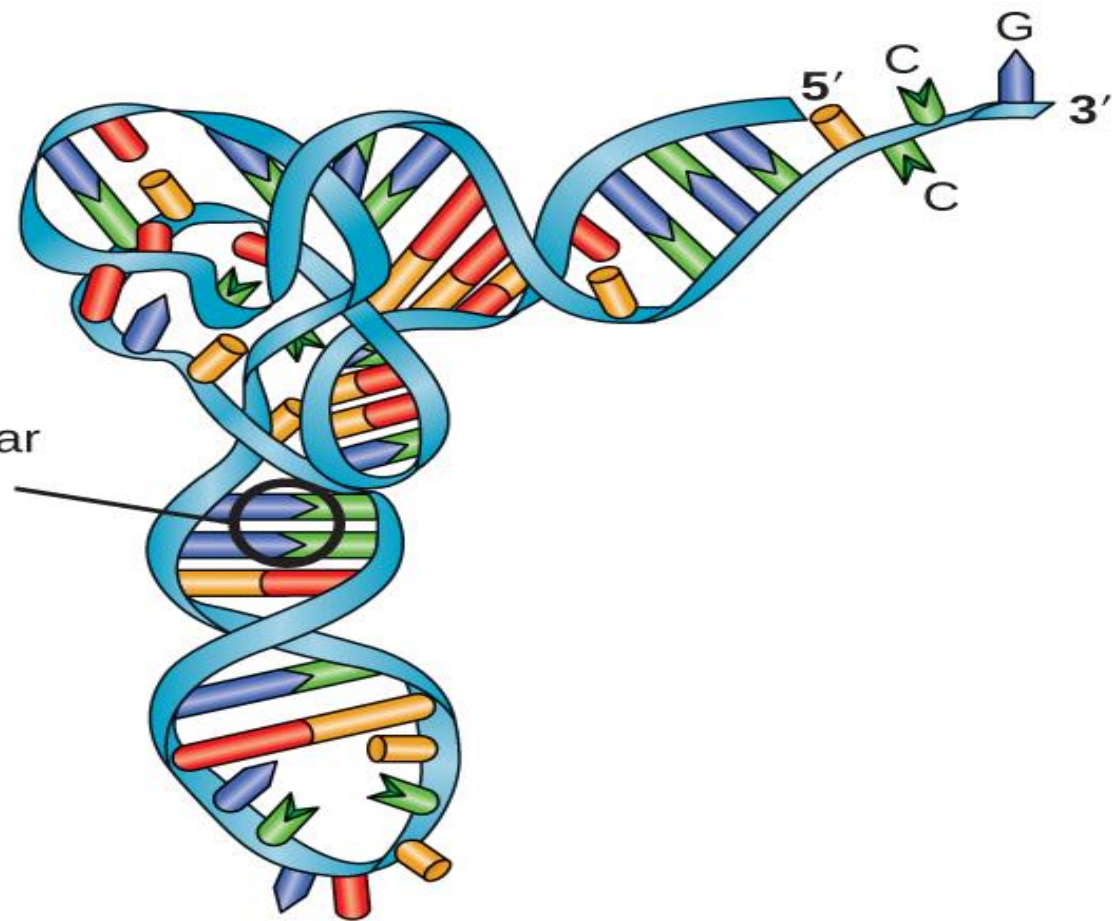
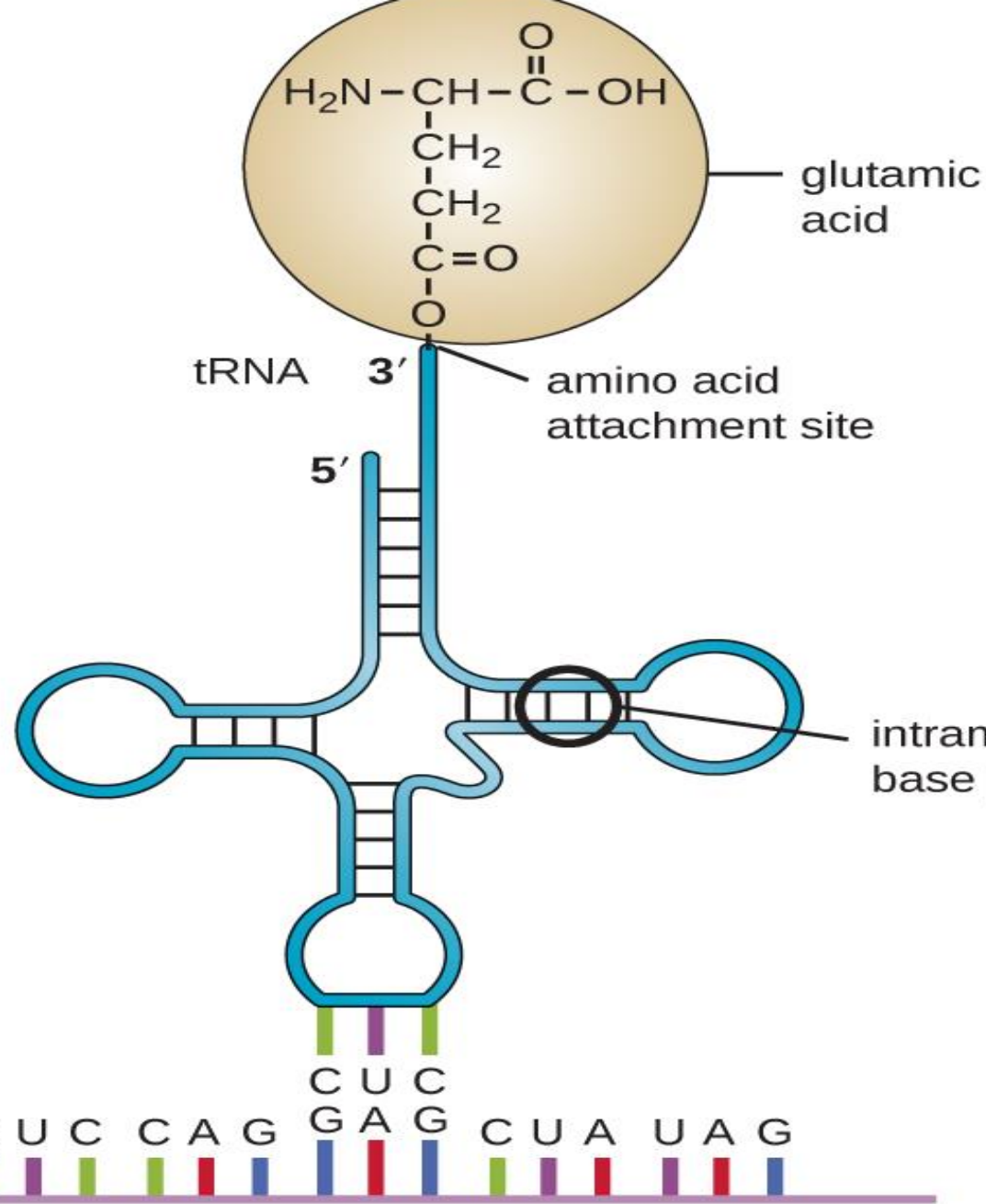
- Ribosomal RNA, as the name suggests, is found in the ***ribosomes***.
- It comprises about 80% of the total RNA of the cell. The base sequence of rRNA is complementary to that of the region of DNA where it is synthesized.
- Ribosomal RNA consists of a single strand twisted upon itself in some regions.
- The ***ribosome*** consists of proteins and RNA.
- The ribosome consists of 2 subunits large and small.



Transfer RNA – tRNA or Soluble RNA – sRNA

- The L-shaped tRNA is formed from a small single-stranded RNA molecule that folds into the proper conformation. Four different regions of double-stranded RNA are formed during the folding process.
- The two ends of the molecule form the ***acceptor stem region*** where the amino acid is attached to one end and the other end consists of three anticodon nucleotides. This consist of unpaired sequences of cytosine-cytosine-adenine(C-C-A) at the 3'end.
- ✓ The 3'-OH terminal of adenine may bind with alpha –COOH of specific amino acid as (aminocyl-tRNA complex).
- ✓ The starting amino acid in eukaryote protein synthesis is methionine, while in prokaryotes it is N-formyl methionine.

- ✓ The tRNA molecules specific for these two amino acids are methionyl tRNA (tRNA^{Met}) and N-formyl-methionyl tRNA (tRNA^{fMet}) respectively. These tRNAs are called initiator tRNAs, because they initiate protein synthesis.
- The anticodon is an exposed single-stranded region in a loop at the end of the ***anticodon arm***. This is another unpaired and non bounded loop carrying specific sequences of 3 bases including the anticodon.
- The anticodon pairs with a codon in messenger RNA (mRNA) ensuring that the correct amino acid is incorporated into the growing polypeptide chain.



mRNA

tRNA 2-dimensional structure

tRNA 3-dimensional structure

More RNAs molecules

- The enzyme *telomerase*, which adds nucleotides to the ends of chromosomes, is composed of a 451-nucleotide RNA and several proteins.
- Other classes of RNA species include *microRNAs*, *small interfering RNAs*, and *sRNAs*—all of which are not translated into proteins but still perform important regulatory functions in the cell.
- The discovery of these RNAs has been one of the most exciting advances in recent years, and there is currently a lot of interest in the use of these molecules as possible therapies.
- These RNAs all share the same basic single-stranded chemical structure with, in some cases, higher-order structures obtained through complementary base-pair folding.